



PRACTICE
QUESTIONS - 01

Python Programming
Level: Basic

DATA SCIENCE

PRACTICE QUESTIONS - 01

BASIC QUESTIONS

1. Calculate the multiplication and sum of two numbers

Given two integer numbers, write a Python program to return their product only if the product is equal to or lower than 1000. Otherwise, return their sum

2. Print the Sum of a Current Number and a Previous number

Write Python code to iterate through the first 10 numbers and, in each iteration, print the sum of the current and previous number.

3: Print characters present at an even index number

Write a Python code to accept a string from the user and display characters present at an even index number.

For example, str = "Intellentx". so your code should display t,l,e,t

4: Remove first n characters from a string

Write a Python code to remove characters from a string from 0 to n and return a new string.

5: Check if the first and last numbers of a list are the same

Write a code to return True if the list's first and last numbers are the same. If the numbers are different, return False.

6: Display numbers divisible by 5

Write a Python code to display numbers from a list divisible by 5

7. Find the number of occurrences of a substring in a string

Write a Python code to find how often the substring "Emma" appears in the given string.

8. Check Palindrome Number

Write a Python code to check if the given number is a palindrome. A palindrome number reads the same forwards and backward. For example, **545 is a palindrome number**.

9. Merge two lists using the following condition

Given two lists of numbers, write Python code to create a new list containing odd numbers from the first list and even numbers from the second list.

10. Get each digit from a number in the reverse order.

For example, If the given integer number is **7536**, the output shall be **"6 3 5 7"**, with a space separating the digits.

11. Print multiplication table from 1 to 10

The multiplication table from 1 to 10 is a table that shows the products of numbers from 1 to 10. Write a code to generates a complete **multiplication table for numbers 1 through 10**.

PRACTICE QUESTIONS - 01

BASIC QUESTIONS

12. Get an int value of base raises to the power of exponent

Write a function called `exponent(base, exp)` that returns an int value of base raises to the power of `exp`.

13. Generate Fibonacci series up to 15 terms

Have you ever wondered about the Fibonacci Sequence? It's a series of numbers in which the next number is found by adding up the two numbers before it. The first two numbers are 0 and 1.

For example, 0, 1, 1, 2, 3, 5, 8, 13, 21. The next number in this series is $13 + 21 = 34$

14. Check if a given year is a leap year

A **leap year** is a year in the Gregorian calendar that contains an extra day, making it 366 days long instead of the usual 365. This extra day, **February 29th**, is added to keep the calendar synchronized with the Earth's revolution around the Sun.

Rules for leap years: a year is a leap year if it's divisible by 4, unless it's also divisible by 100 but not by 400.

15. Print Alternate Prime Numbers till 20

A **Prime Number** is a number that can only be divided by itself and 1 without remainders (e.g., 2, 3, 5, 7, 11).